

CODE : CSA0311

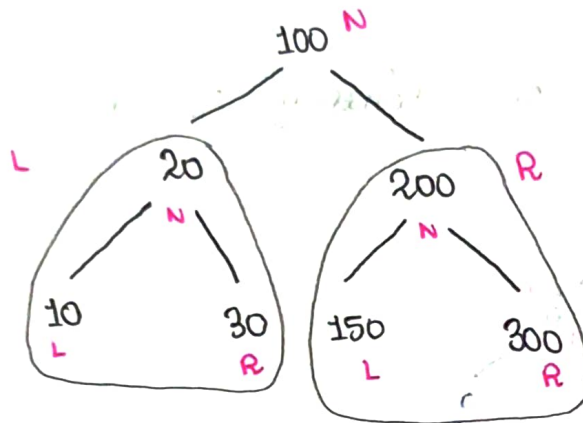
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COURSE NAME : Data Structure

REG NO: 192421285

Co3 AT3 Tree Traversal Challenges

①



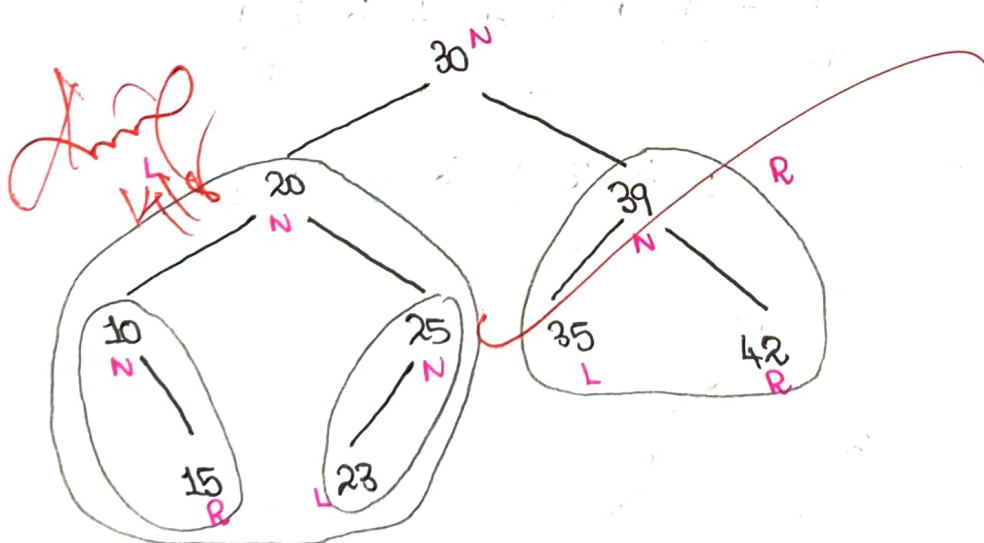
Inorder : LNR $\rightarrow$ $\frac{10 \quad 20 \quad 30}{L} \quad \frac{100}{N} \quad \frac{150 \quad 800 \quad 900}{R}$

Preorder: NLR $\rightarrow$ $\frac{100}{N}$ $\frac{20 \quad 10 \quad 30}{L}$ $\frac{200 \quad 150 \quad 300}{R}$

Postorden: LRN $\rightarrow$

10	30	20	150	300	200	100
L			R			N

②

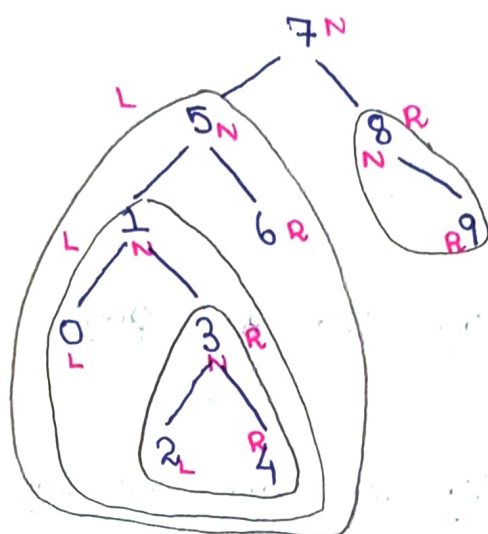


Inorder : LNR → 10 15 20 23 25 30 35 40 42
 L N R

Preorder: NLR $\rightarrow$ $\frac{30}{N}$ $\frac{20}{L}$ $\frac{10}{L}$ $\frac{15}{L}$ $\frac{25}{L}$ $\frac{23}{L}$ $\frac{39}{R}$ $\frac{35}{R}$ $\frac{42}{R}$

Postorder: LRN $\rightarrow$ $\frac{15}{L}$ $\frac{10}{L}$ $\frac{23}{L}$ $\frac{25}{L}$ $\frac{20}{L}$ $\frac{35}{R}$ $\frac{42}{R}$ $\frac{39}{R}$ $\frac{30}{N}$

③ 7 5 1 8 3 6 0 9 4 2 $\rightarrow$ Binary Search Tree

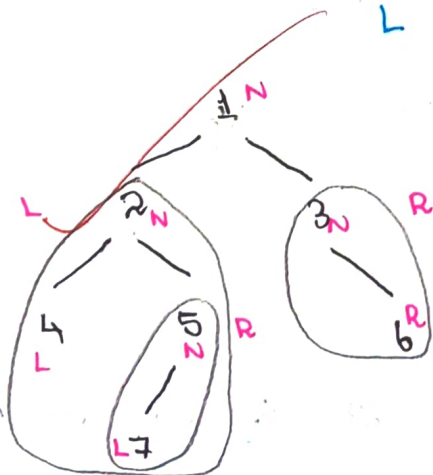


Inorder: LNR $\rightarrow$ $\frac{0}{L}$ $\frac{1}{L}$ $\frac{2}{L}$ $\frac{3}{L}$ $\frac{4}{L}$ $\frac{5}{L}$ $\frac{6}{L}$ $\frac{7}{N}$ $\frac{8}{R}$ $\frac{9}{R}$

Preorder: NLR $\rightarrow$ $\frac{7}{N}$ $\frac{5}{L}$ $\frac{1}{L}$ $\frac{0}{L}$ $\frac{3}{L}$ $\frac{2}{L}$ $\frac{4}{L}$ $\frac{6}{L}$ $\frac{8}{R}$ $\frac{9}{R}$

Postorder: LRN $\rightarrow$ $\frac{0}{L}$ $\frac{2}{L}$ $\frac{4}{L}$ $\frac{3}{L}$ $\frac{1}{L}$ $\frac{6}{L}$ $\frac{5}{L}$ $\frac{9}{R}$ $\frac{8}{R}$ $\frac{7}{N}$

⑤

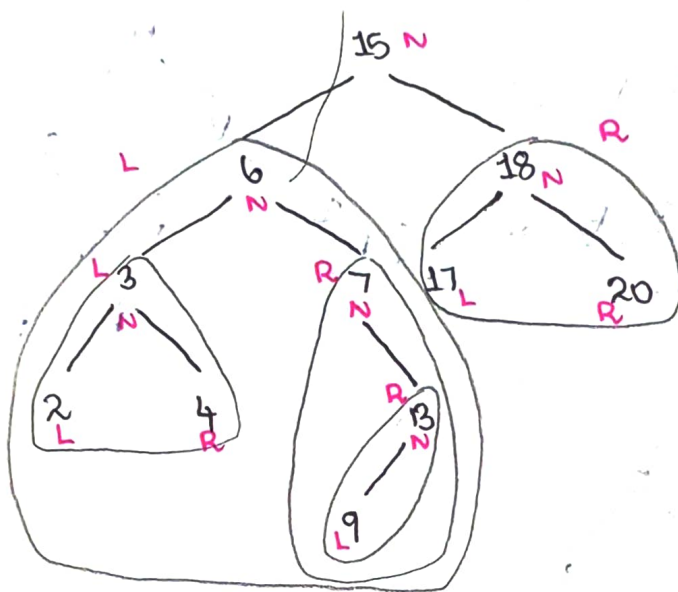


Inorder: LNR $\rightarrow$ $\frac{7 \ 4 \ 2 \ 5}{L} \quad \frac{1}{N} \quad \frac{3 \ 6}{R}$

Preorder: NLR $\rightarrow$ $\frac{1}{N} \quad \frac{2 \ 4 \ 7 \ 5}{L} \quad \frac{3 \ 6}{R}$

Postorder: LRN $\rightarrow$ $\frac{7 \ 4 \ 5 \ 2}{L} \quad \frac{6 \ 3}{R} \quad \frac{1}{N}$

⑥

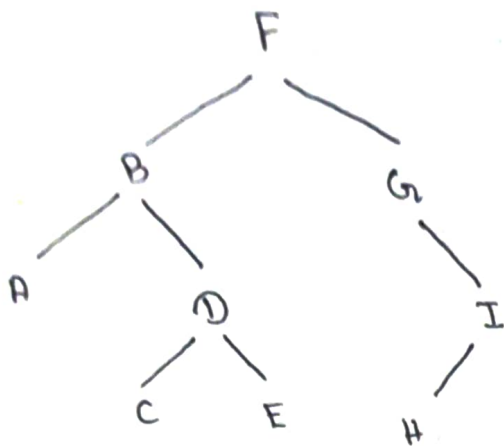


Inorder: LNR $\rightarrow$ $\frac{2 \ 3 \ 4 \ 6 \ 7 \ 9 \ 13}{L} \quad \frac{15}{N} \quad \frac{17 \ 18 \ 20}{R}$

Preorder: NLR $\rightarrow$ $\frac{15}{N} \quad \frac{6 \ 3 \ 2 \ 4 \ 7 \ 13 \ 9}{L} \quad \frac{18 \ 17 \ 20}{R}$

Postorder: LRN $\rightarrow$ $\frac{2 \ 4 \ 3 \ 9 \ 13 \ 7 \ 6}{L} \quad \frac{17 \ 20 \ 18}{R} \quad \frac{15}{N}$

⑦



Inorder: LNR

A B C D E F G H I
 L N R

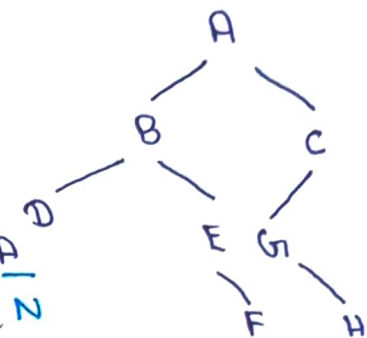
⑧

Inorder - D B E F A G H C

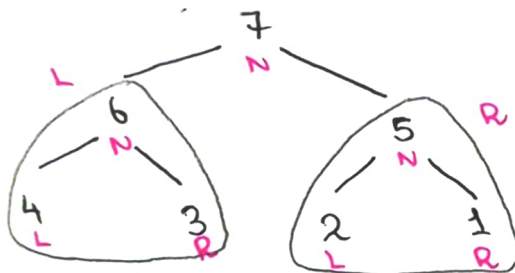
Preorder - A B D E F C G H

Postorder: LRN →

D F E B G H C A
 L R N



⑨



Inorder :- LNR →

4 6 3 2 5 1
 L R
 7
 N

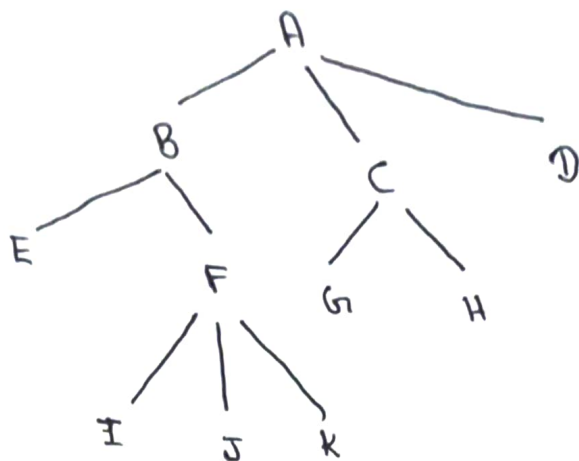
Preorder: NLR →

7 6 4 3 5 2 1
 L R
 N

Postorder: LRN →

4 3 6 2 1 5 7
 L R N

10



Inorder: LNR $\rightarrow$ Not defined

Preorder: NLR $\rightarrow$ A B E F I J K C G H D

Postorder: LRN $\rightarrow$ E I J K F B G H C D A